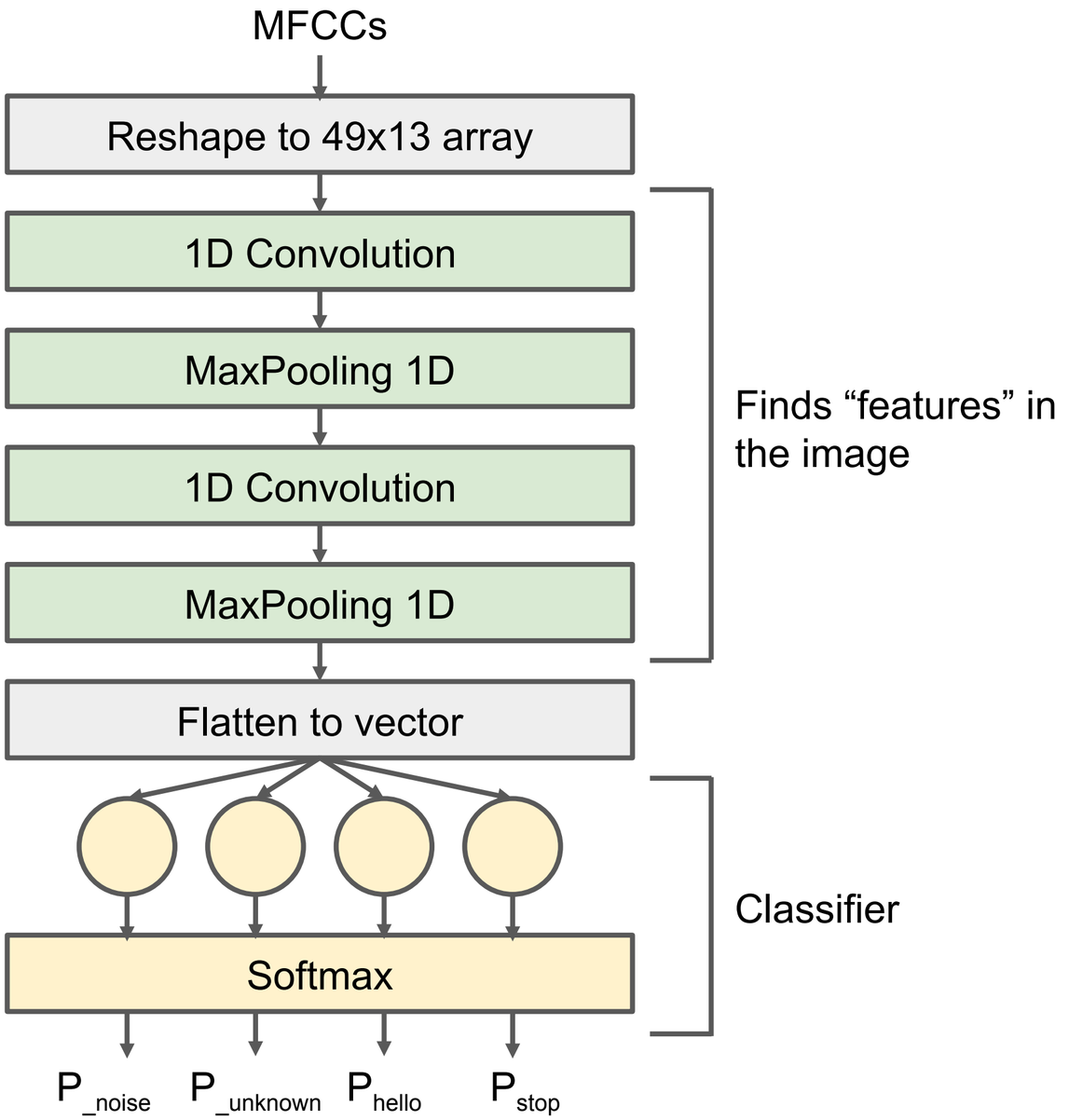
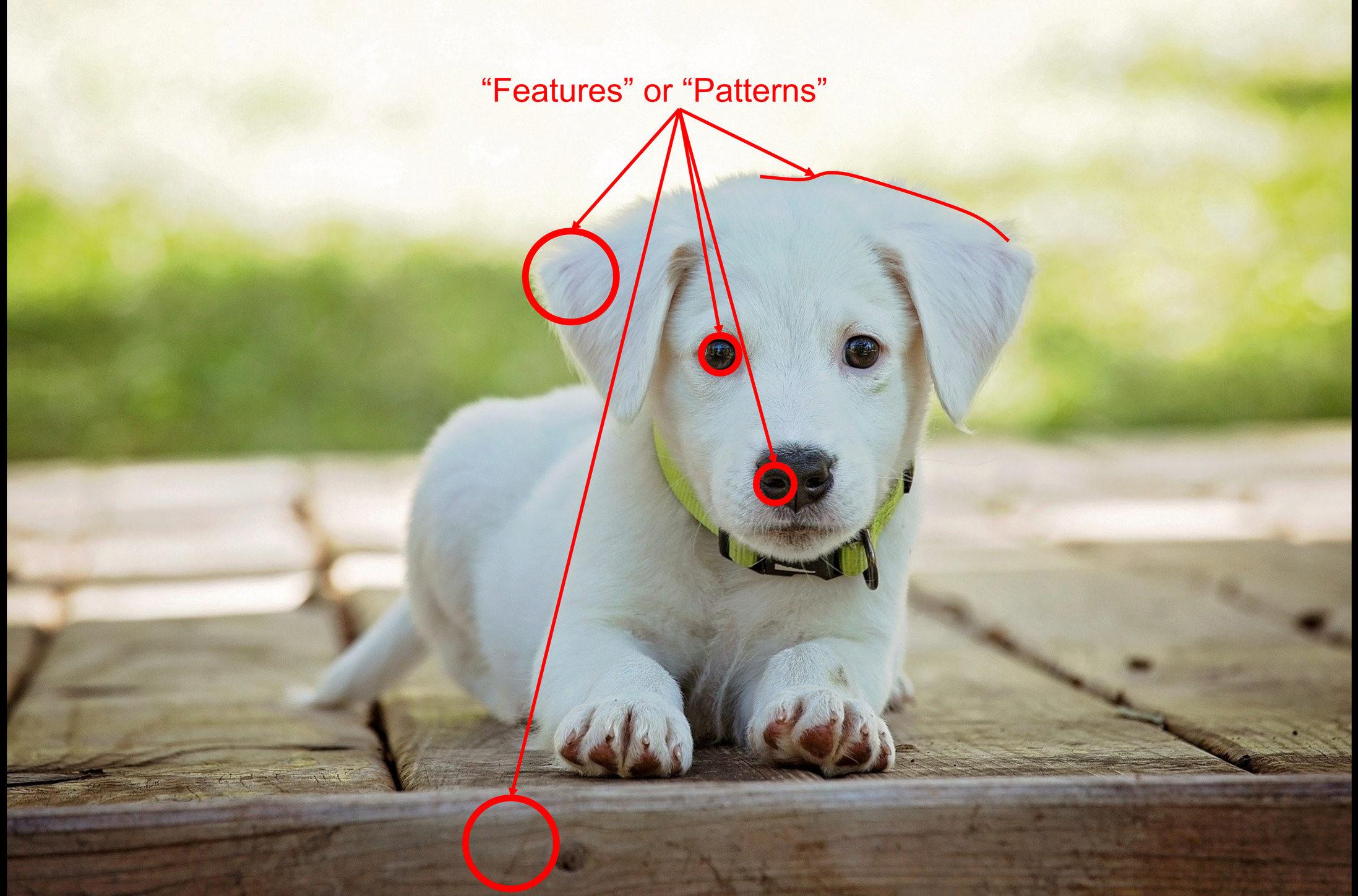
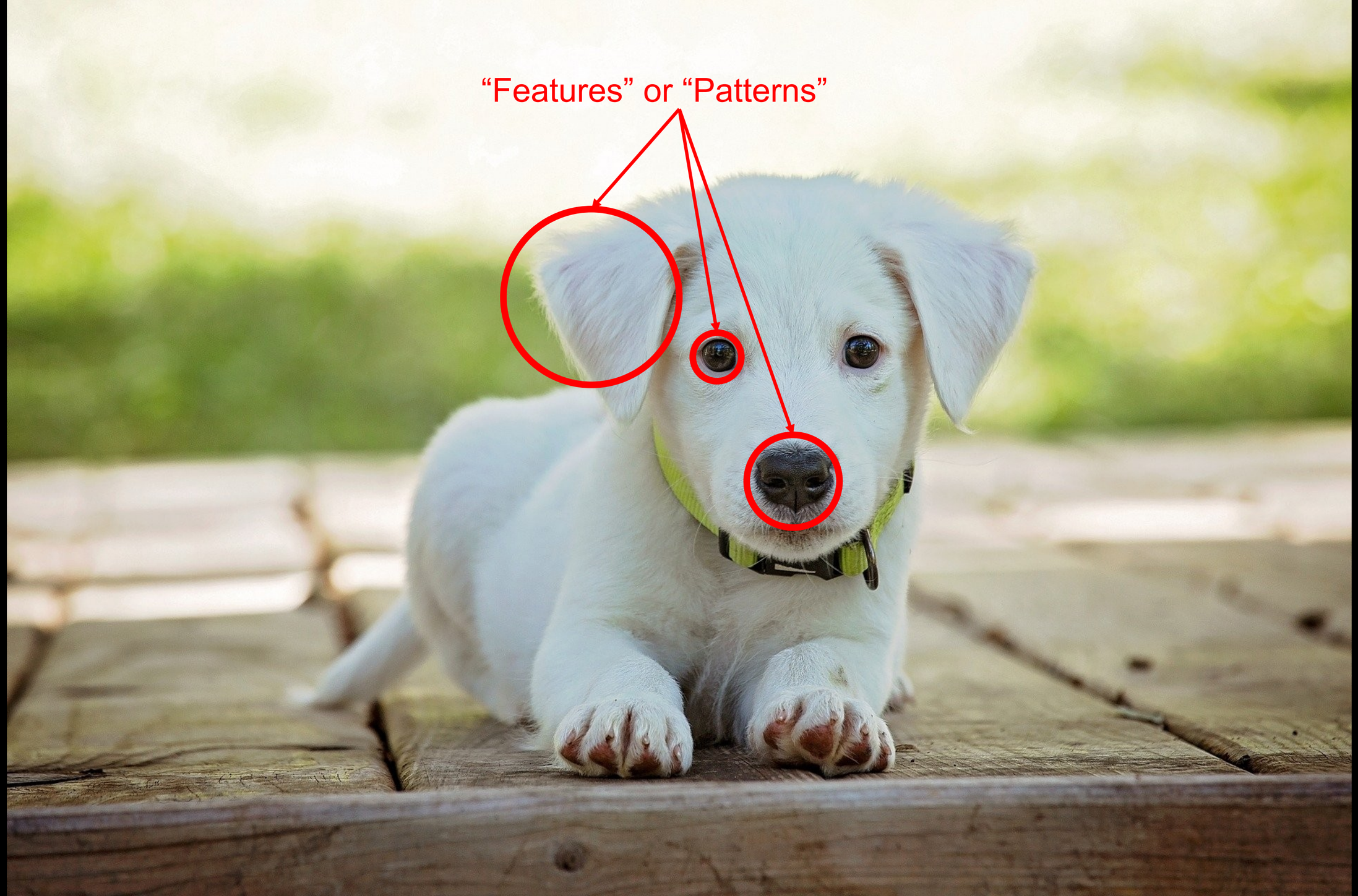
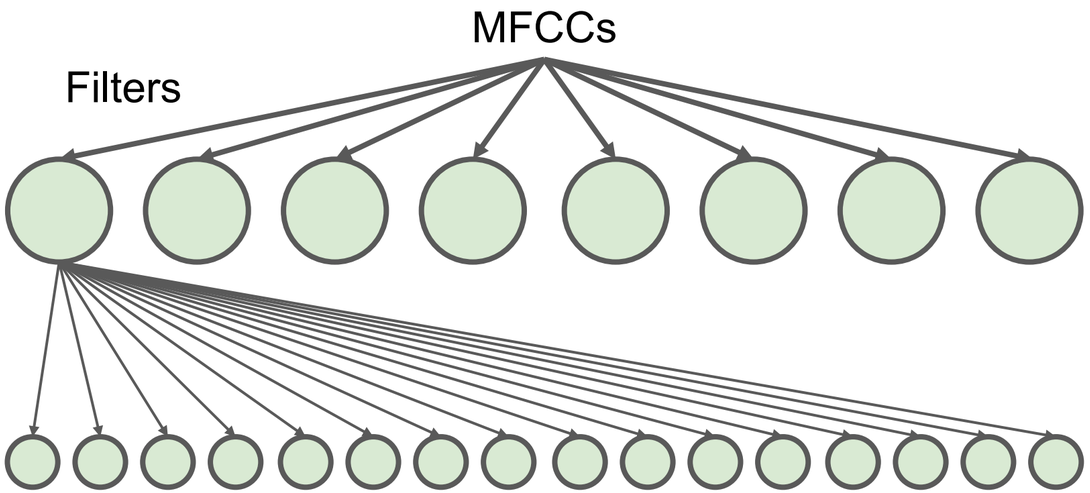
Convolutional Neural Network (CNN or ConvNet)

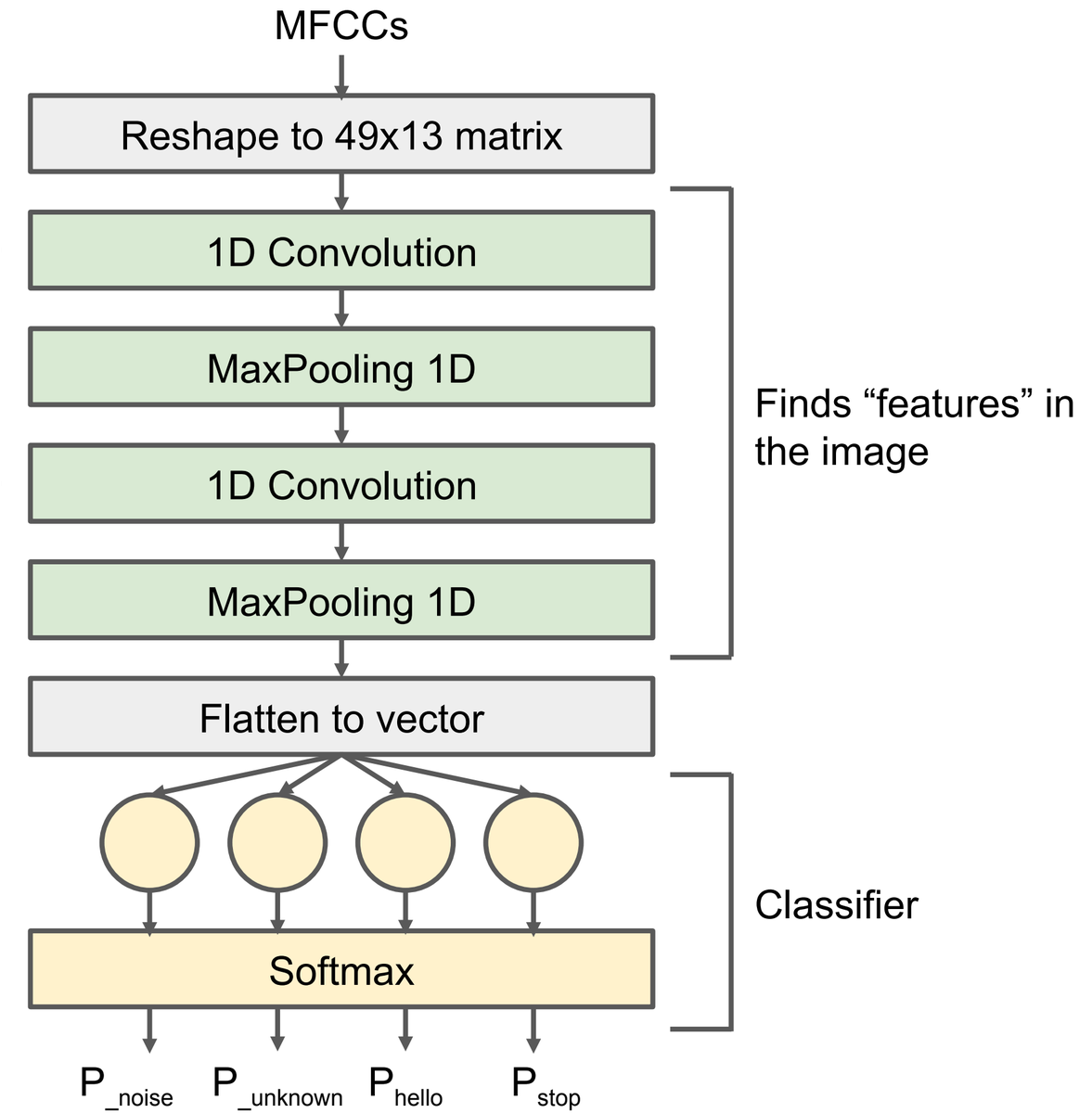



“Features” or “Patterns”


“Features” or “Patterns”

Convolutional Neural Network (CNN or ConvNet)

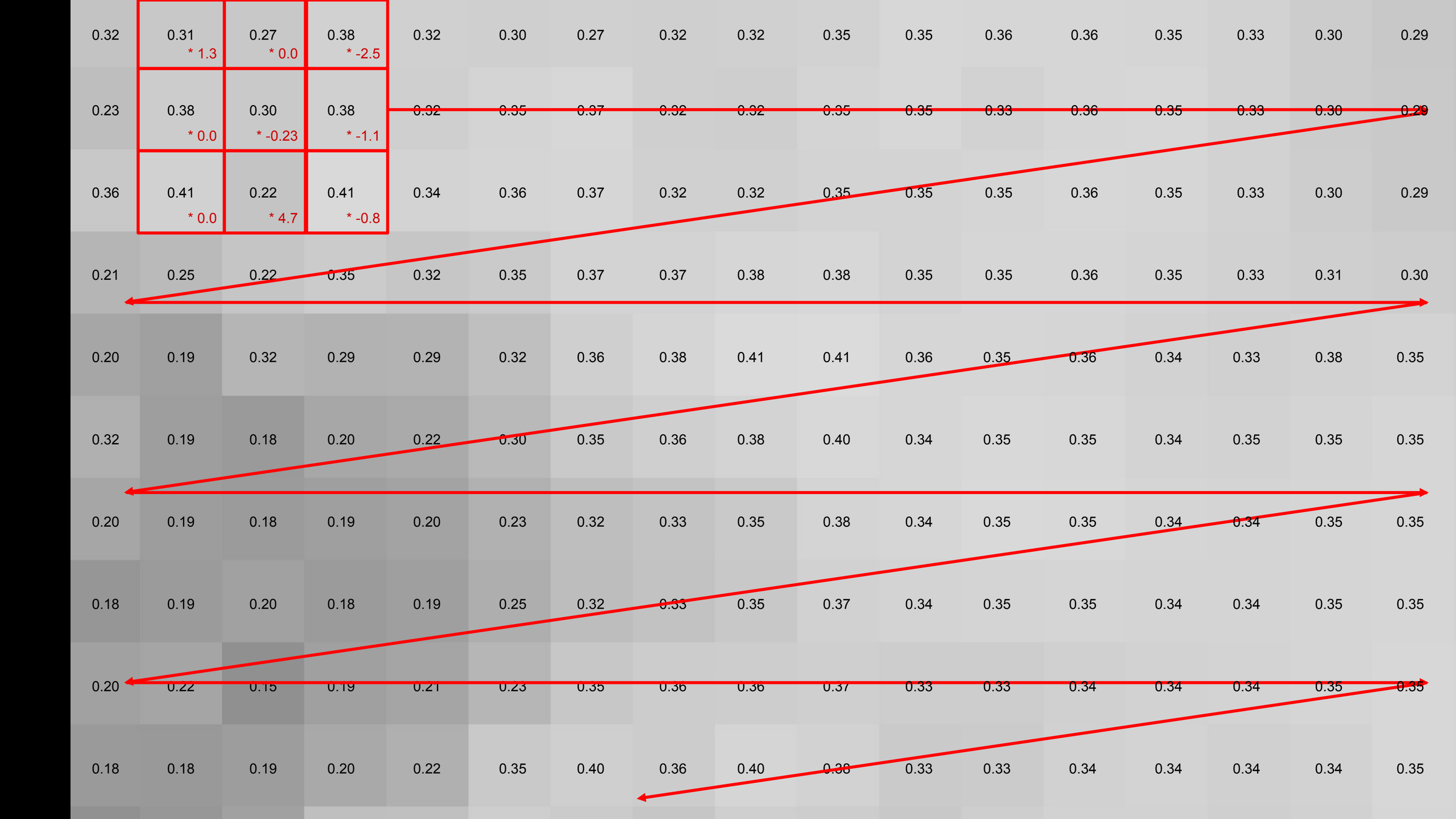


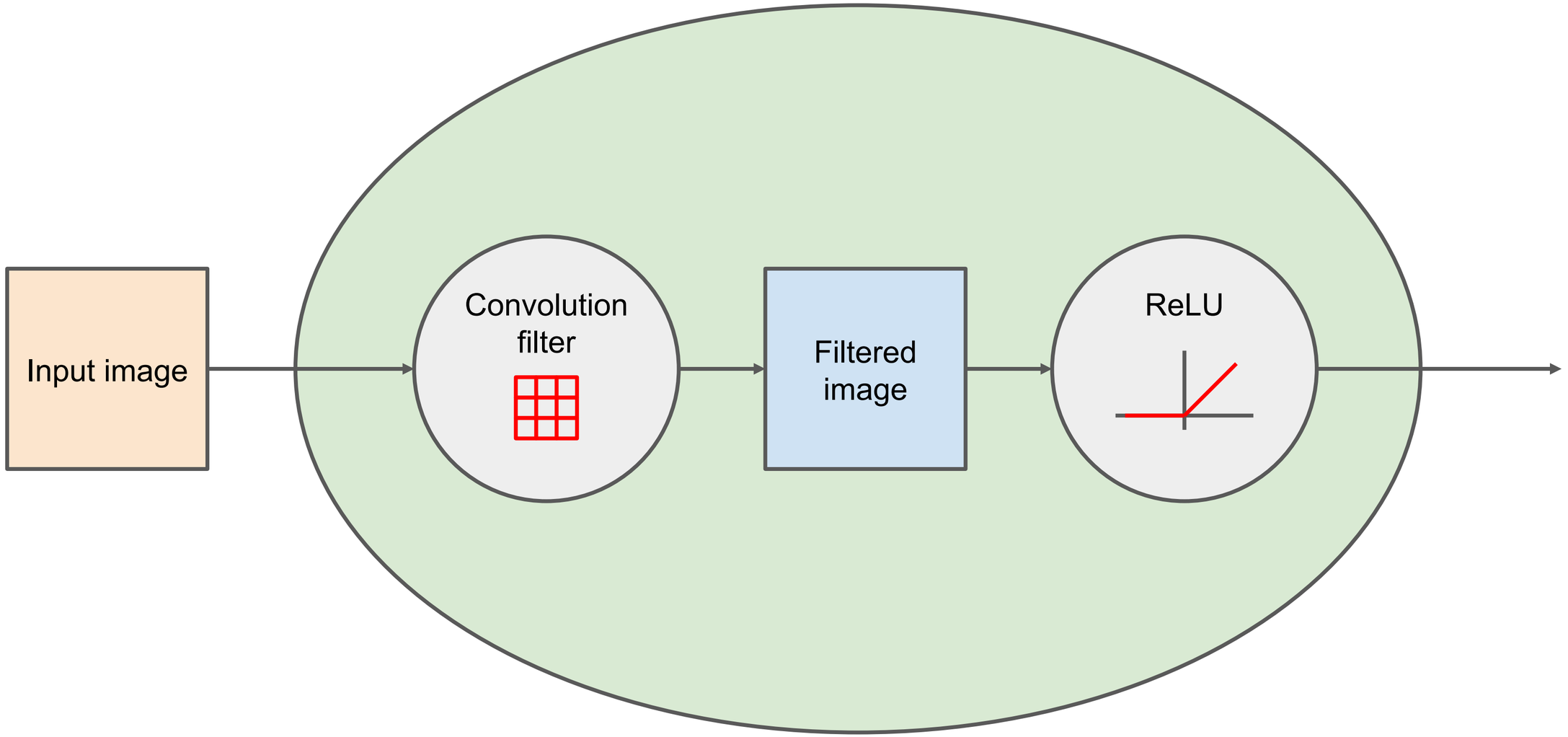






0.32 * 1.3	0.31 * 0.0	0.27 * -2.5	0.38	0.32	0.30	0.27	0.32	0.32	0.35	0.35	0.36	0.36	0.35	0.33	0.30	0.29
0.23 * 0.0	0.38 * -0.23	0.30 * -1.1	0.38	0.32	0.35	0.37	0.32	0.32	0.35	0.35	0.33	0.36	0.35	0.33	0.30	0.29
0.36 * 0.0	0.41 * 4.7	0.22 * -0.8	0.41	0.34	0.36	0.37	0.32	0.32	0.35	0.35	0.35	0.36	0.35	0.33	0.30	0.29
0.21	0.25	0.22	0.35	0.32	0.35	0.37	0.37	0.38	0.38	0.35	0.35	0.36	0.35	0.33	0.31	0.30
0.20	0.19	0.32	0.29	0.29	0.32	0.36	0.38	0.41	0.41	0.36	0.35	0.36	0.34	0.33	0.38	0.35
0.32	0.19	0.18	0.20	0.22	0.30	0.35	0.36	0.38	0.40	0.34	0.35	0.35	0.34	0.35	0.35	0.35
0.20	0.19	0.18	0.19	0.20	0.23	0.32	0.33	0.35	0.38	0.34	0.35	0.35	0.34	0.34	0.35	0.35
0.18	0.19	0.20	0.18	0.19	0.25	0.32	0.33	0.35	0.37	0.34	0.35	0.35	0.34	0.34	0.35	0.35
0.20	0.22	0.15	0.19	0.21	0.23	0.35	0.36	0.36	0.37	0.33	0.33	0.34	0.34	0.34	0.35	0.35
0.18	0.18	0.19	0.20	0.22	0.35	0.40	0.36	0.40	0.38	0.33	0.33	0.34	0.34	0.34	0.34	0.35


0.32 0.31 * 1.3 0.27 * 0.0 0.38 * -2.5 0.32 0.30 0.27 0.32 0.32 0.35 0.35 0.36 0.36 0.35 0.33 0.30 0.29
0.23 0.38 * 0.0 0.30 * -0.23 0.38 * -1.1 0.32 0.35 0.37 0.32 0.32 0.35 0.35 0.33 0.36 0.35 0.33 0.30 0.29
0.36 0.41 * 0.0 0.22 * 4.7 0.41 * -0.8 0.34 0.36 0.37 0.32 0.32 0.35 0.35 0.35 0.36 0.35 0.33 0.30 0.29
0.21 0.25 0.22 0.35 0.32 0.35 0.37 0.37 0.38 0.38 0.35 0.35 0.36 0.35 0.33 0.31 0.30
0.20 0.19 0.32 0.29 0.29 0.32 0.36 0.38 0.41 0.41 0.36 0.35 0.36 0.34 0.33 0.38 0.35
0.32 0.19 0.18 0.20 0.22 0.30 0.35 0.36 0.38 0.40 0.34 0.35 0.35 0.34 0.35 0.35 0.35
0.20 0.19 0.18 0.19 0.20 0.23 0.32 0.33 0.35 0.38 0.34 0.35 0.35 0.34 0.34 0.35 0.35
0.18 0.19 0.20 0.18 0.19 0.25 0.32 0.33 0.35 0.37 0.34 0.35 0.35 0.34 0.34 0.35 0.35
0.20 0.22 0.15 0.19 0.21 0.23 0.35 0.36 0.36 0.37 0.33 0.33 0.34 0.34 0.34 0.35 0.35
0.18 0.18 0.19 0.20 0.22 0.35 0.40 0.36 0.40 0.38 0.33 0.33 0.34 0.34 0.34 0.34 0.35



Single node in convolution layer

1D Convolution

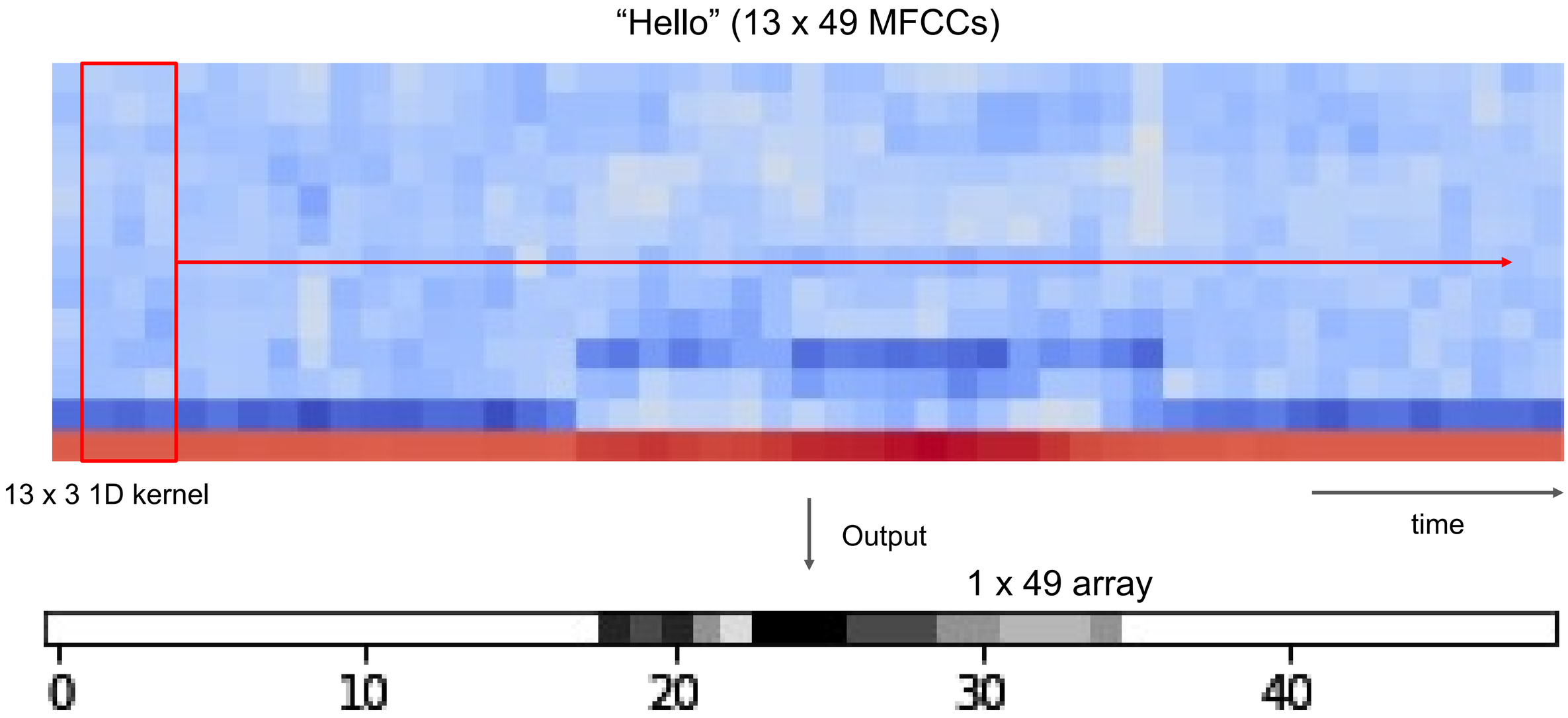
“hello”



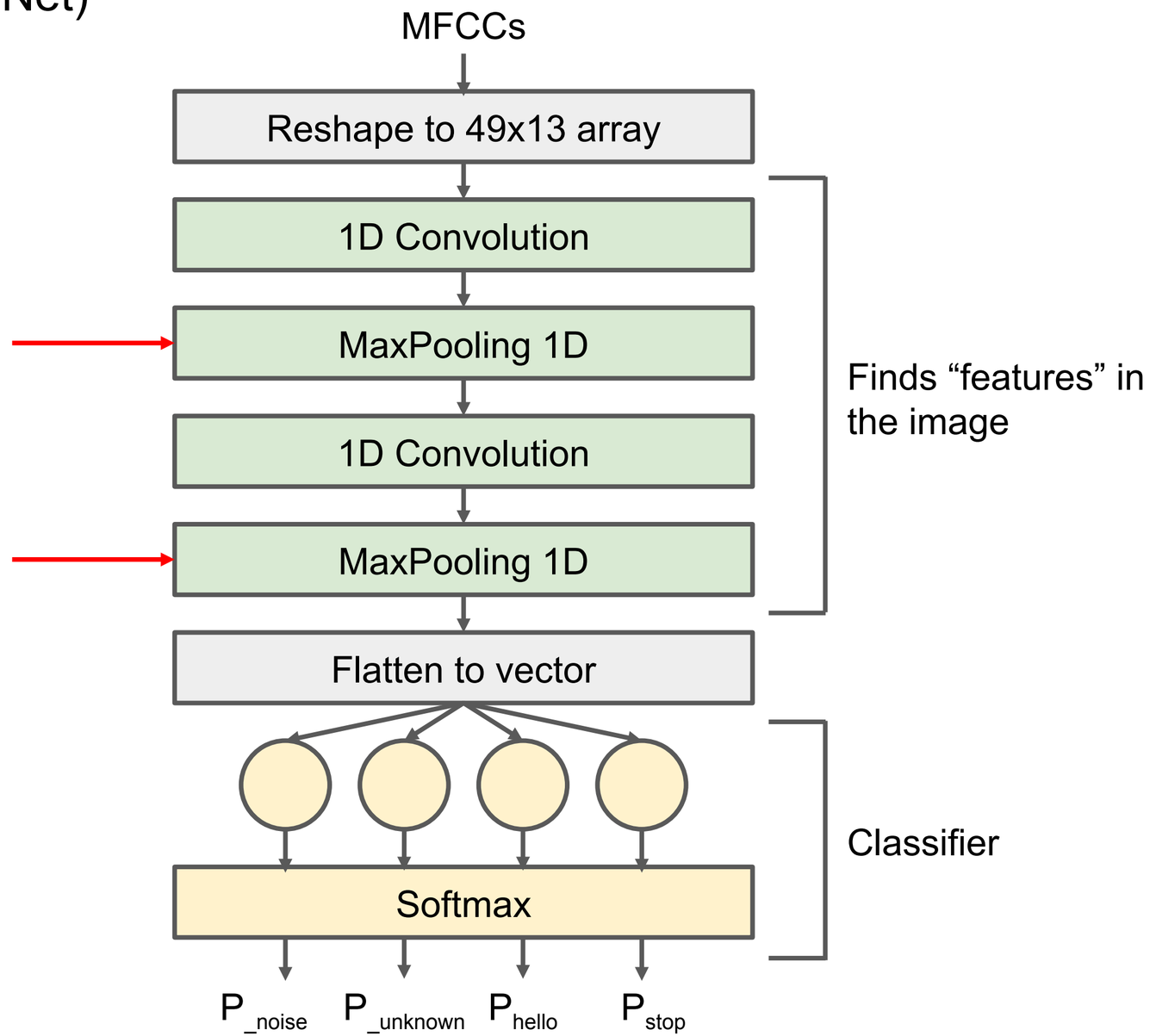
3 x 13 1D kernel

time

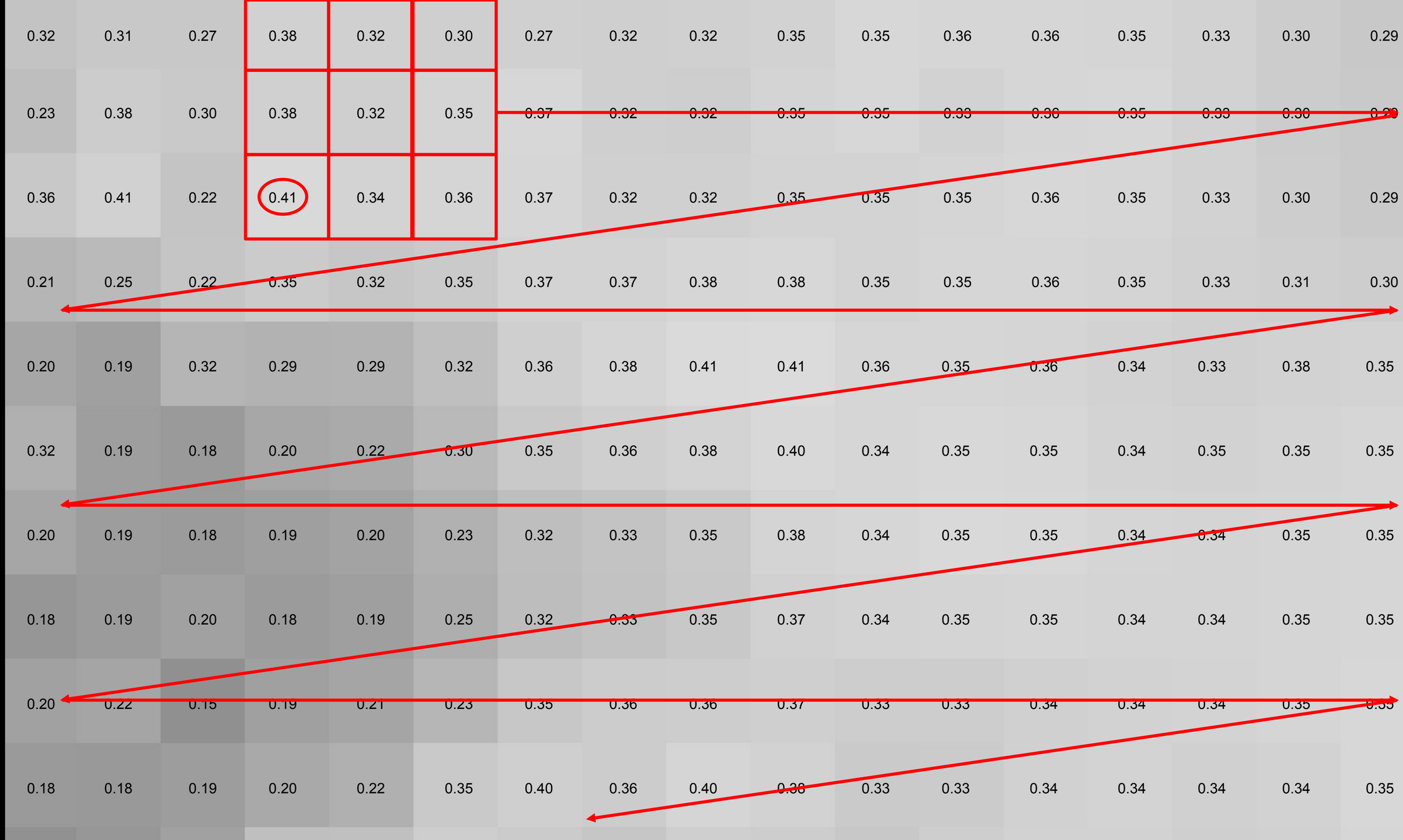
1D Convolution



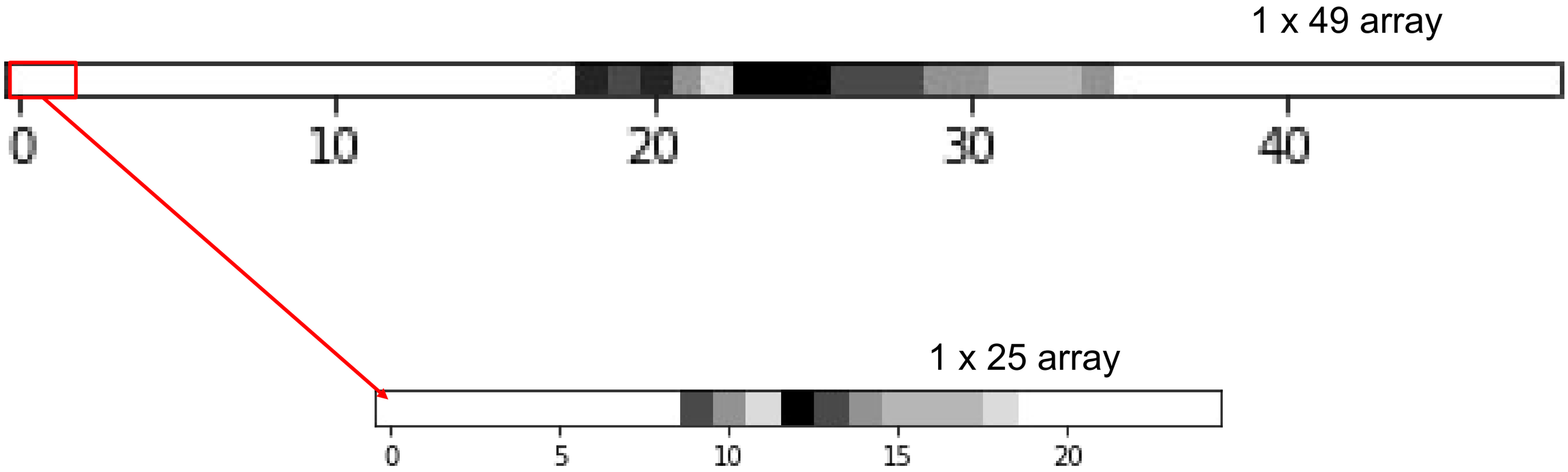
Convolutional Neural Network (CNN or ConvNet)



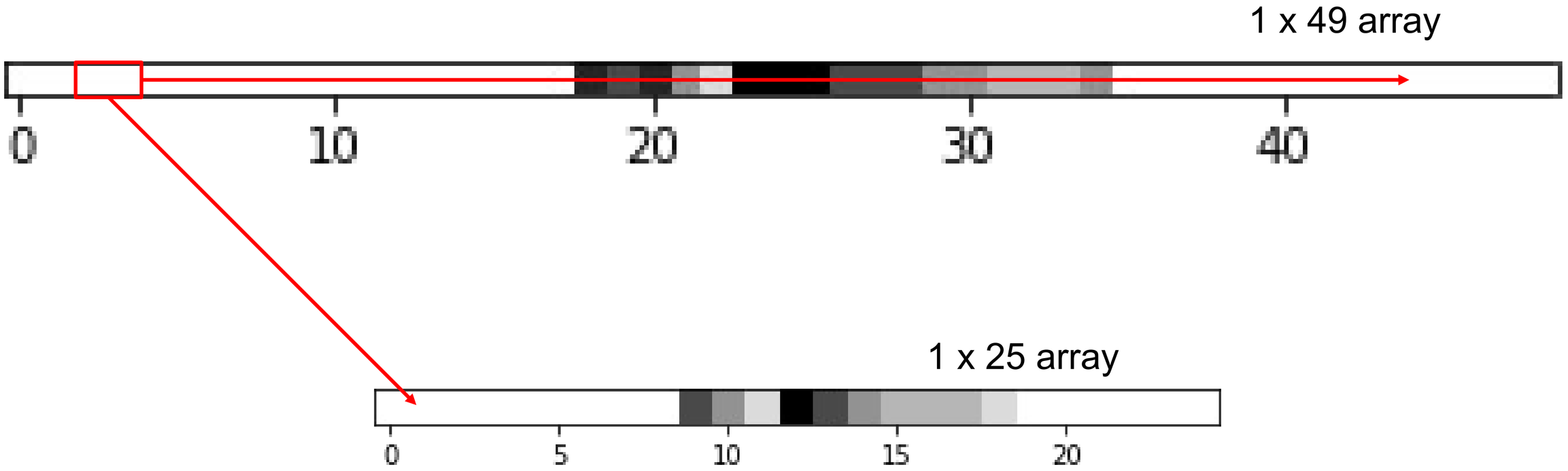
0.32	0.31	0.27	0.38	0.32	0.30	0.27	0.32	0.32	0.35	0.35	0.36	0.36	0.35	0.33	0.30	0.29
0.23	0.38	0.30	0.38	0.32	0.35	0.37	0.32	0.32	0.35	0.35	0.33	0.36	0.35	0.33	0.30	0.29
0.36	0.41	0.22	0.41	0.34	0.36	0.37	0.32	0.32	0.35	0.35	0.35	0.36	0.35	0.33	0.30	0.29
0.21	0.25	0.22	0.35	0.32	0.35	0.37	0.37	0.38	0.38	0.35	0.35	0.36	0.35	0.33	0.31	0.30
0.20	0.19	0.32	0.29	0.29	0.32	0.36	0.38	0.41	0.41	0.36	0.35	0.36	0.34	0.33	0.38	0.35
0.32	0.19	0.18	0.20	0.22	0.30	0.35	0.36	0.38	0.40	0.34	0.35	0.35	0.34	0.35	0.35	0.35
0.20	0.19	0.18	0.19	0.20	0.23	0.32	0.33	0.35	0.38	0.34	0.35	0.35	0.34	0.34	0.35	0.35
0.18	0.19	0.20	0.18	0.19	0.25	0.32	0.33	0.35	0.37	0.34	0.35	0.35	0.34	0.34	0.35	0.35
0.20	0.22	0.15	0.19	0.21	0.23	0.35	0.36	0.36	0.37	0.33	0.33	0.34	0.34	0.34	0.35	0.35
0.18	0.18	0.19	0.20	0.22	0.35	0.40	0.36	0.40	0.38	0.33	0.33	0.34	0.34	0.34	0.34	0.35


0.32 0.31 0.27 0.38 0.32 0.30 0.27 0.32 0.32 0.35 0.35 0.36 0.36 0.35 0.33 0.30 0.29
0.23 0.38 0.30 0.38 0.32 0.35 0.37 0.32 0.32 0.35 0.35 0.33 0.36 0.35 0.33 0.30 0.29
0.36 0.41 0.22 0.41 0.34 0.36 0.37 0.32 0.32 0.35 0.35 0.35 0.36 0.35 0.33 0.30 0.29
0.21 0.25 0.22 0.35 0.32 0.35 0.37 0.37 0.38 0.38 0.35 0.35 0.36 0.35 0.33 0.31 0.30
0.20 0.19 0.32 0.29 0.29 0.32 0.36 0.38 0.41 0.41 0.36 0.35 0.36 0.34 0.33 0.38 0.35
0.32 0.19 0.18 0.20 0.22 0.30 0.35 0.36 0.38 0.40 0.34 0.35 0.35 0.34 0.35 0.35 0.35
0.20 0.19 0.18 0.19 0.20 0.23 0.32 0.33 0.35 0.38 0.34 0.35 0.35 0.34 0.34 0.35 0.35
0.18 0.19 0.20 0.18 0.19 0.25 0.32 0.33 0.35 0.37 0.34 0.35 0.35 0.34 0.34 0.35 0.35
0.20 0.22 0.15 0.19 0.21 0.23 0.35 0.36 0.36 0.37 0.33 0.33 0.34 0.34 0.34 0.35 0.35
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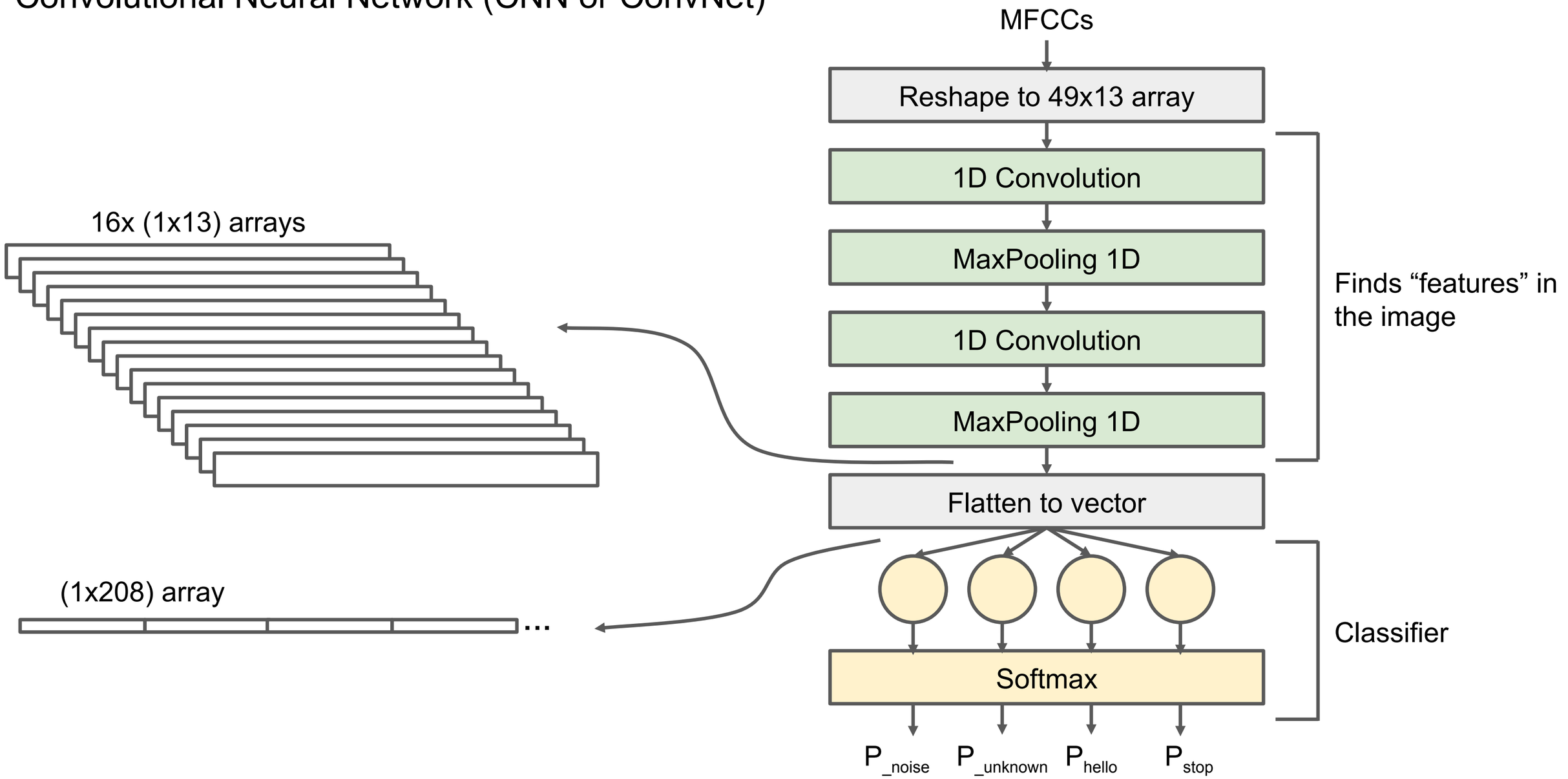
1D MaxPooling



1D MaxPooling



Convolutional Neural Network (CNN or ConvNet)



Convolutional Neural Network (CNN or ConvNet)

